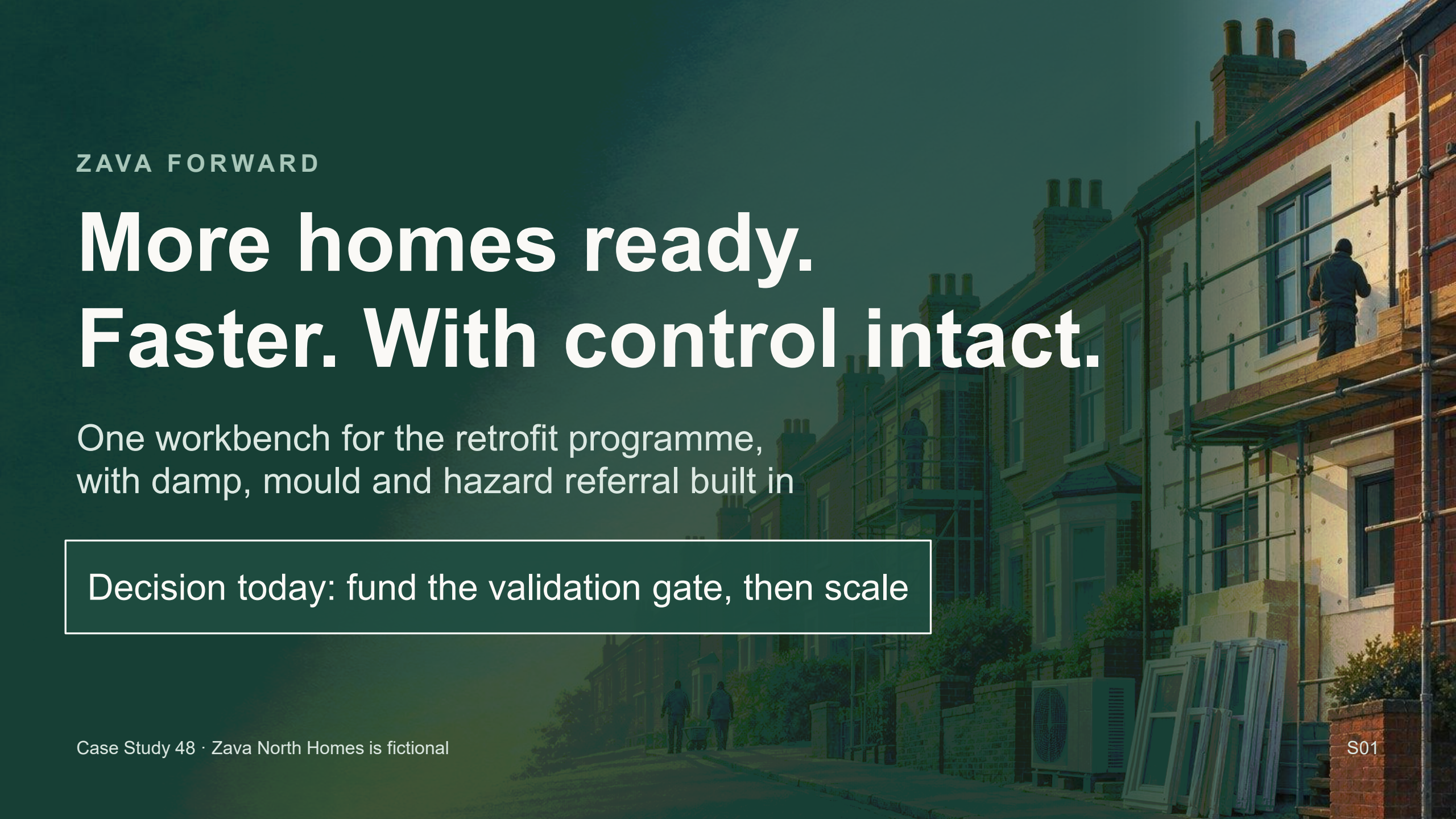


ZAVA FORWARD

More homes ready. Faster. With control intact.

One workbench for the retrofit programme, with damp, mould and hazard referral built in

Decision today: fund the validation gate, then scale

ACT 1 · THE CHALLENGE

15,600 homes to lift by 2030. Thousands more to keep safe.

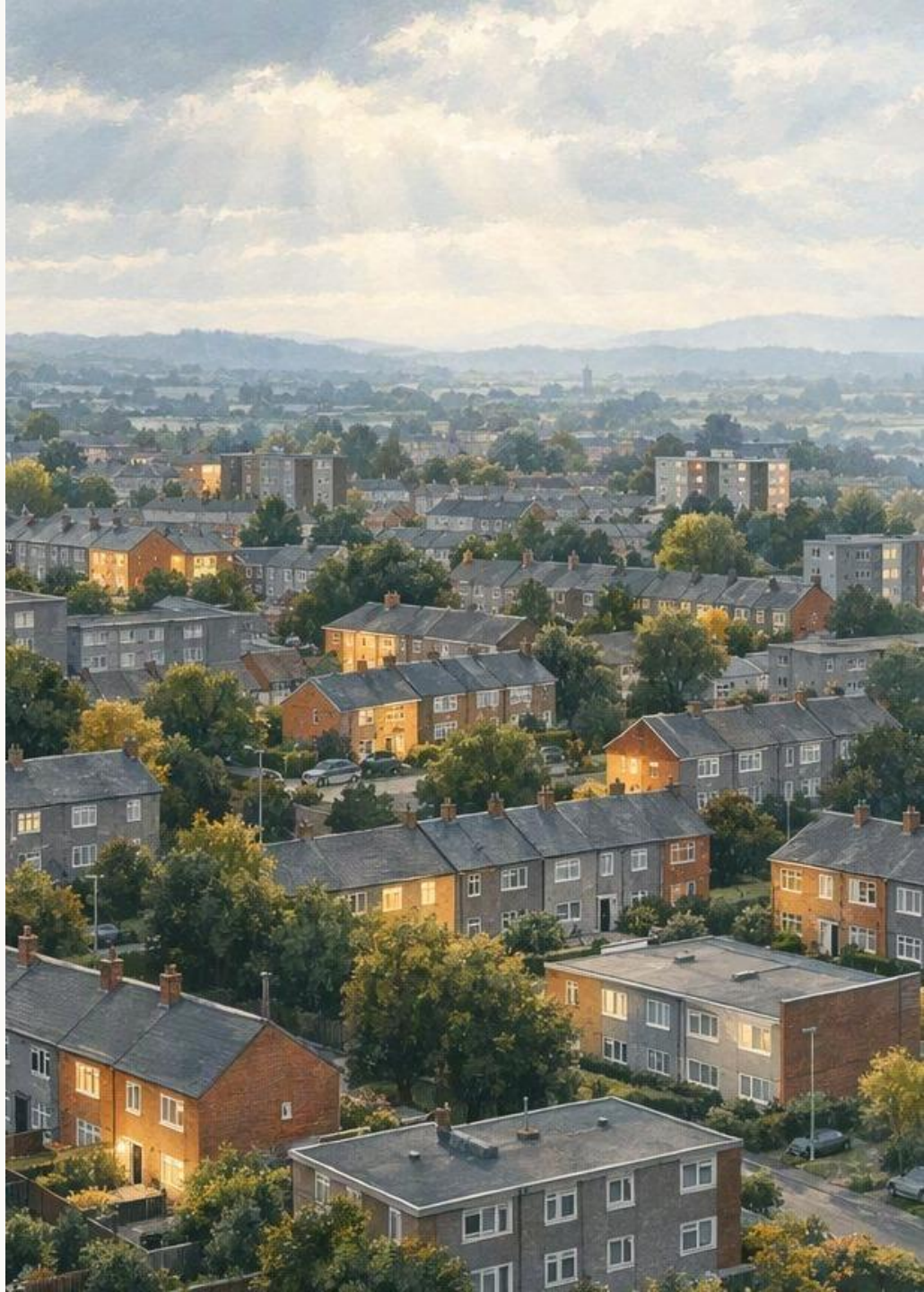
15,600
homes below EPC C
26% of social homes

3,600
homes with damp
6% of housing association homes

22,800
households likely to report condensation, damp or mould
38%

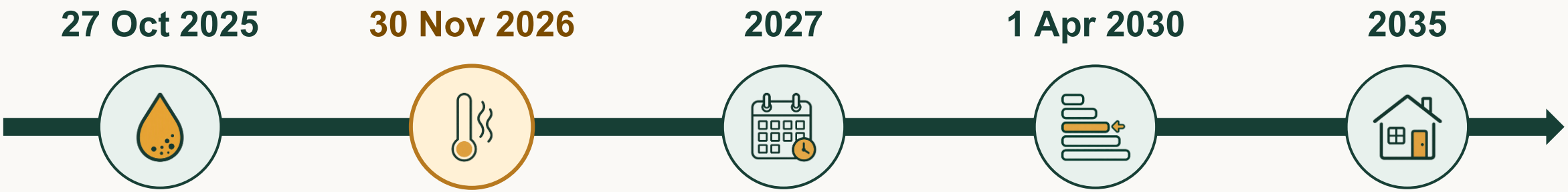
3,210
stage-one complaints a year
53.5 per 1,000 homes

Sources: English Housing Survey 2024-25 · RSH Tenant Satisfaction Measures 2024/25 · national rates applied to 60,000 homes



ACT 1 · THE CHALLENGE

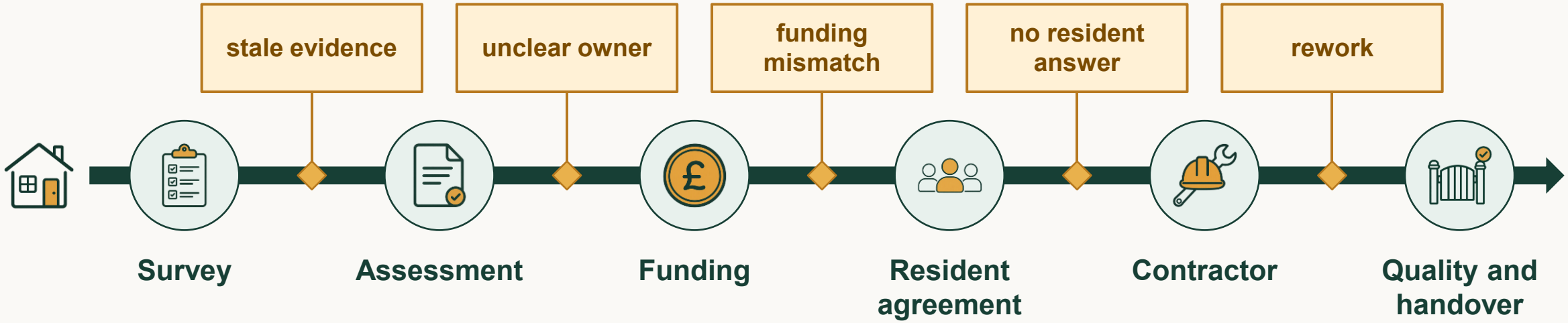
Three clocks are already running.



From 30 Nov 2026, cold homes are a legal hazard: retrofit and repairs become one programme.

Sources: MHCLG Awaab's Law guidance (Jul 2026) · energy efficiency response (Apr 2026) · Decent Homes policy (Jan 2026)

Homes don't stall on ambition. They stall between teams.



Evidence, funding, capacity and resident choice must align

More starts are not the same as more accepted homes

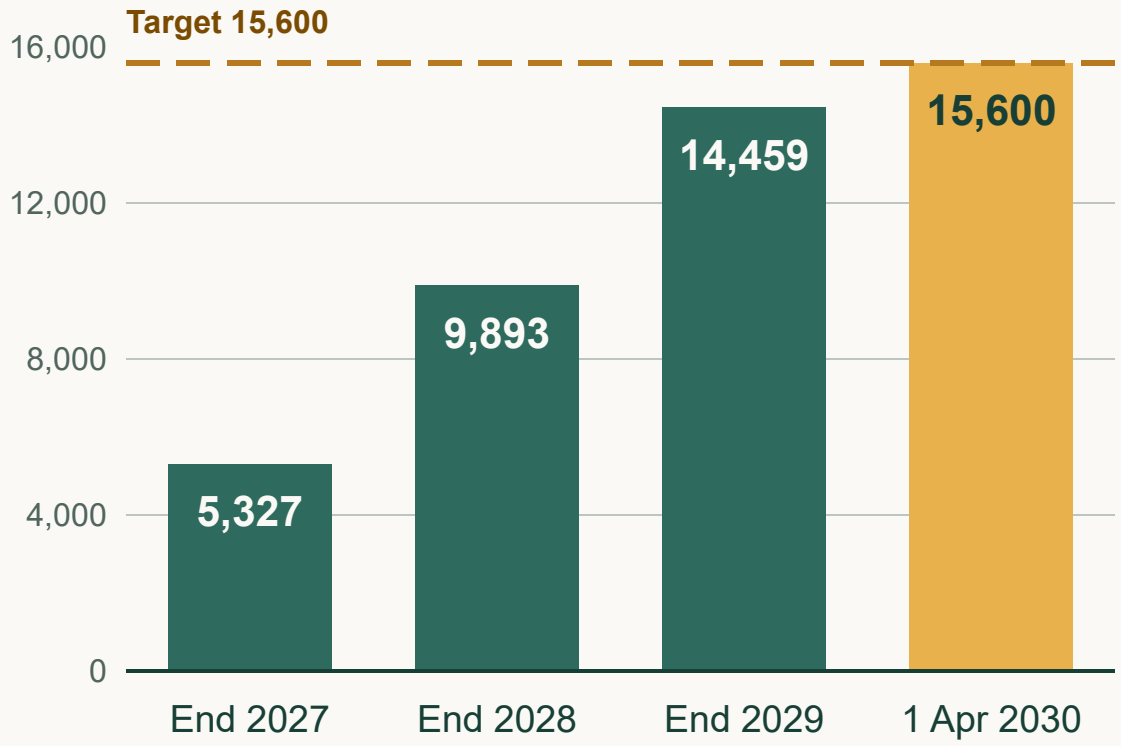
A conflicting record should stop one approval, not all useful work

Sources: exercise journeys, role research and agent-request challenge

Forecast: every Zava home at EPC C by April 2030.

FORECAST

Homes at EPC C, cumulative from Nov 2026



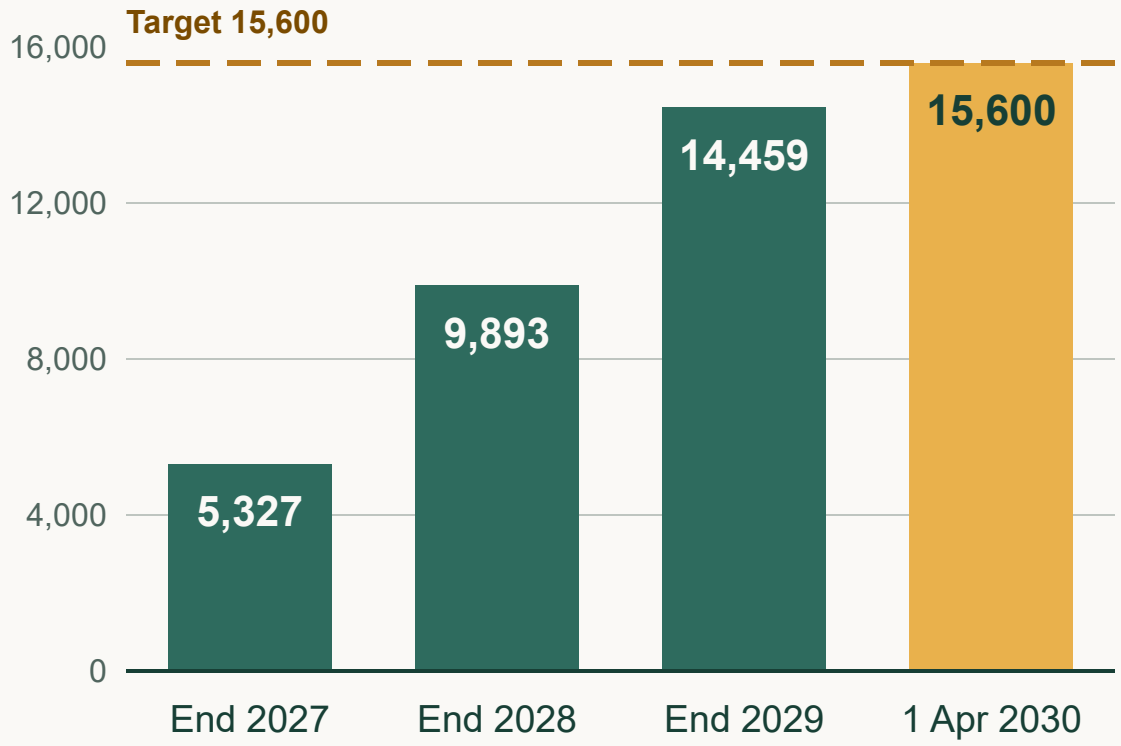
15,600 homes · 41 months

88 homes a week

Forecast: every Zava home at EPC C by April 2030.

FORECAST

Homes at EPC C, cumulative from Nov 2026



15,600 homes · 41 months

88 homes a week

GBP 95m
works benchmark

GBP 47.4m
conditional grant potential

USD 6,873
whole-cohort inference scenario

Forecast: 15,600 homes; E21: assumed workload + secondary price (23 Oct 2026). Grant subject to eligibility, allocation, award and claim.



ACT 2 · DISCOVERY

Six weeks of product work sit under these slides.

What you see: this deck

480 commits	**233** merged pull requests	**6** architecture decisions	**142** requirements
196 acceptance criteria	**34** research documents	**28** journey maps	**189** backlog items

Since 11 Sep 2026
one architect, GitHub Copilot and automation

Sources: public repository jmservera/ama at 140d8f4 (23 Oct 2026) · BRD v0.19.0 · PRD v0.18.0 · AMA Plan board

We designed for the people who move a home forward.

9 role contributions / 28 stage-based maps / 24 bounded opportunities

Desk research + role instruments, not resident or staff validation

1 Report / nominate | 2 Understand home | 3 Decide scope | 4 Coordinate ready work | 5 Deliver / capture | 6 Inspect / remedy | 7 Handover / live with it | 8 Reconcile / learn

1-2 / PAIN HYPOTHESIS

Repeated story

TEST: one sourced case;
safe sharing, less repetition

4 / PAIN HYPOTHESIS

Unowned promises

TEST: named receiver;
sender chases acceptance

3 + 6 / PAIN HYPOTHESIS

Stale reviews

TEST: separate decisions;
reopen affected reviews

Journey atlas: 14 Oct research / 23 Oct reconciliation. Counts describe research coverage, not measured benefit.

ACT 2 · DISCOVERY

We challenged the brief to make it shippable.

BRIEF SAID → Five autonomous agents, autonomy L2-L4

WE BUILT

Five bounded assistants on one proposal service

Why: our 9 role instruments concluded none of the five should run end to end.

BRIEF SAID → Fix the spreadsheets

WE BUILT

Data ownership and workflow instrumentation first, then AI

Why: slow delivery is a coordination problem that spreadsheets hide.

BRIEF SAID → AI makes decisions

WE BUILT

AI prepares, people decide

Why: consent, safety and money need a named human.

BRIEF SAID → Resident portal

UNDER ASSESSMENT

Optional portal; assisted routes stay

Test fewer calls and staff interruptions (#441).

Sources: agent-request challenge (E03) · ADR 0002 and ADR 0003 (E06, E08, E09)

S08 · Answers: B03 · B14

Every prototype taught us something we kept.

20 Sep
PR #1 requirements review
open decisions made explicit

6 Oct
review of 11 perspectives, 9 filters, 8 cases
inconsistent owner labels fixed in requirements

11 Oct
PR #58 real AI, Azure SQL and Blob
decision model + generative calls work

23 Oct
PR #433 Tokenomics
AI is managed by cost per accepted outcome



PR #19
four role-based perspectives
roles need their own queues

8 Oct
MVP replaces prototype
one shared workbench

14 Oct
five assistants, real proposals, 73 SQL tests, survived a restart
the proposal service holds

From a requirements review to five real AI assistants in 25 days

Owner feedback reshaped the screen: my cases, next owner, plain status

Every lesson became a requirement, a decision or a test

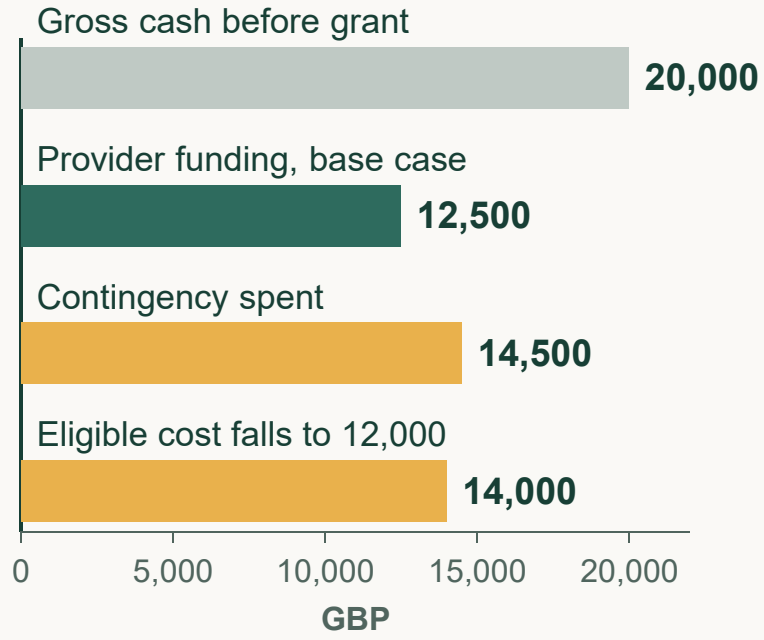
Sources: PRD revision history and learnings · exercise addendum · prototype learnings review · pull requests #1, #19, #58, #433

WHAT-IF

We stress-tested the economics before we wrote the product.

20 scenarios
five agents, a programme supervisor and a damp and mould loop

ONE HOME · exercise fixture (fictional)



THE PROGRAMME · 15,600 homes

WORKS

GBP 6,082/home historical HA benchmark — **94.9m**

By EPC band — **£107m**

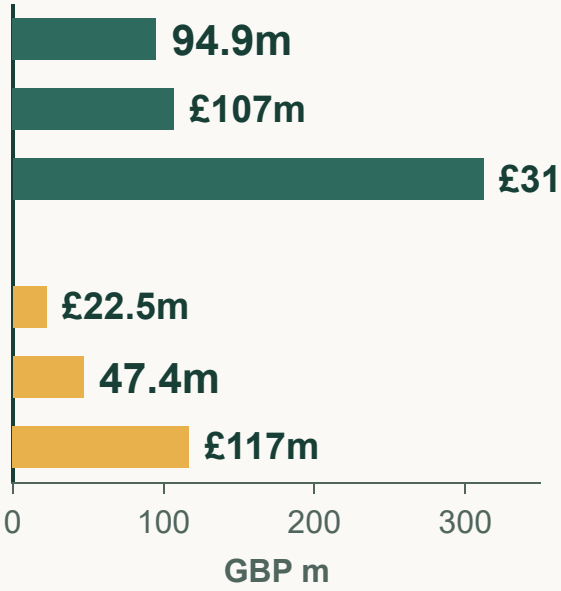
Deep-retrofit fixture (GBP 20,000 a home) — **£312m**

GRANT SCENARIOS

Proportional wave 3 share — **£22.5m**

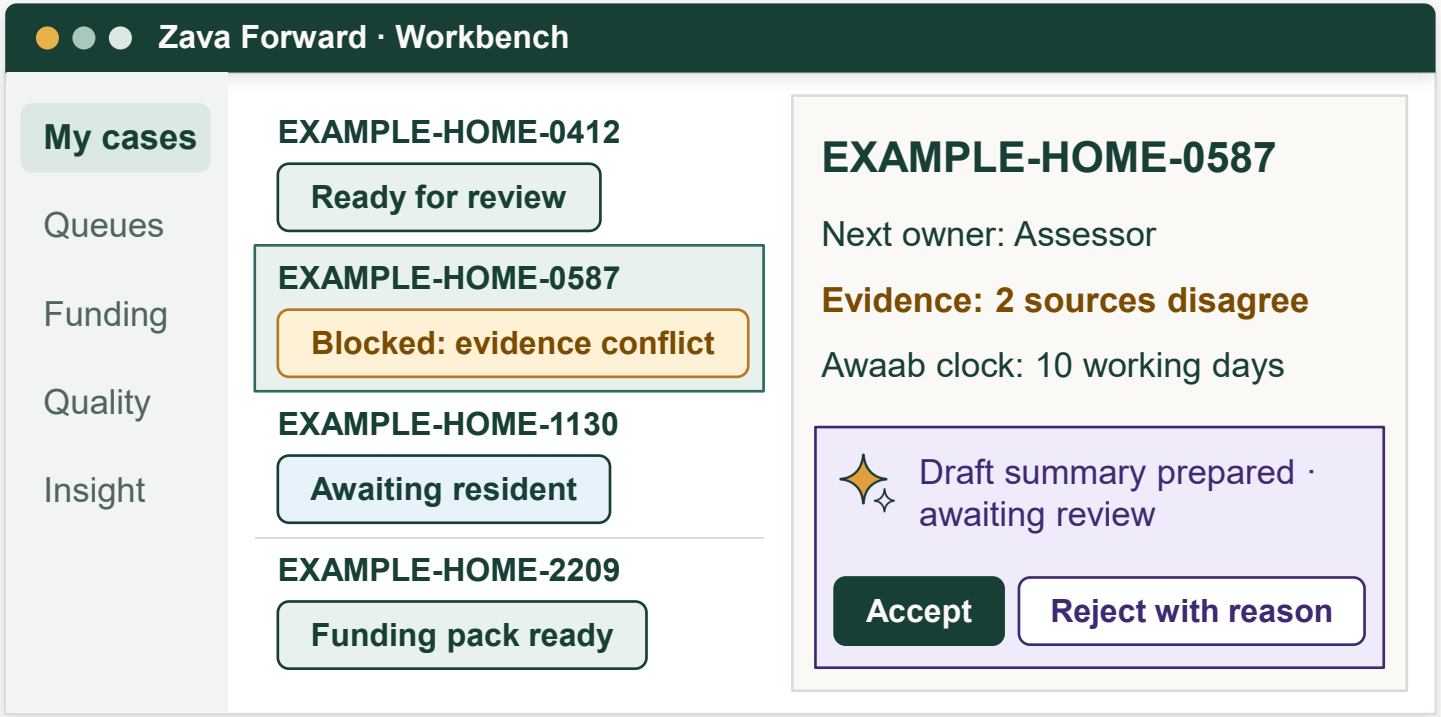
Conditional GBP 3,041/home (50% share) — **47.4m**

Full £7,500 cap — **£117m**



E21: dated benchmark, not a quote. Grant assumes eligible homes/costs, allocation, award and claim. GBP 20,000 fixture remains separate.

Meet the Workbench: every home, one owner, one next step.



My cases · next owner · plain status · bounded help

Delivered in the MVP: work-queue signals, controlled bulk edits, hazard referral, grant states, forecasts

Every decision sits on one evidence trail

Product concept view based on delivered work queue (#77), case controls (#78) and AI suggestions (#106). Fictional data.

Sources: AMA Plan board (23 Oct 2026): #73, #76, #77, #78, #85, #95, #102, #106 · prototype learnings

One home, end to end: every exception has an owner.

Fictional service synthesis: a proposed experience, not an actual system trace or delivered-feature claim

	Report / understand	Choose scope	Book / deliver	Inspect / handover	Claim / reconcile
Resident	Tell it once; choose support	Agree, defer or decline	Confirm slot; raise changes	Practise controls; report concerns	Keep aftercare and redress
Delivery team	Resolve identity; record evidence	**CHANGE: scope review reopens**	Accept capacity; release / capture	Remedy returns; explain handover	Retain defects and open owners
Finance + QA	Check route / evidence gaps	Finance: budget Resident: works choice	**CHANGE: recheck commitment version**	Independent QA; accept or return	Reconcile claim against cash

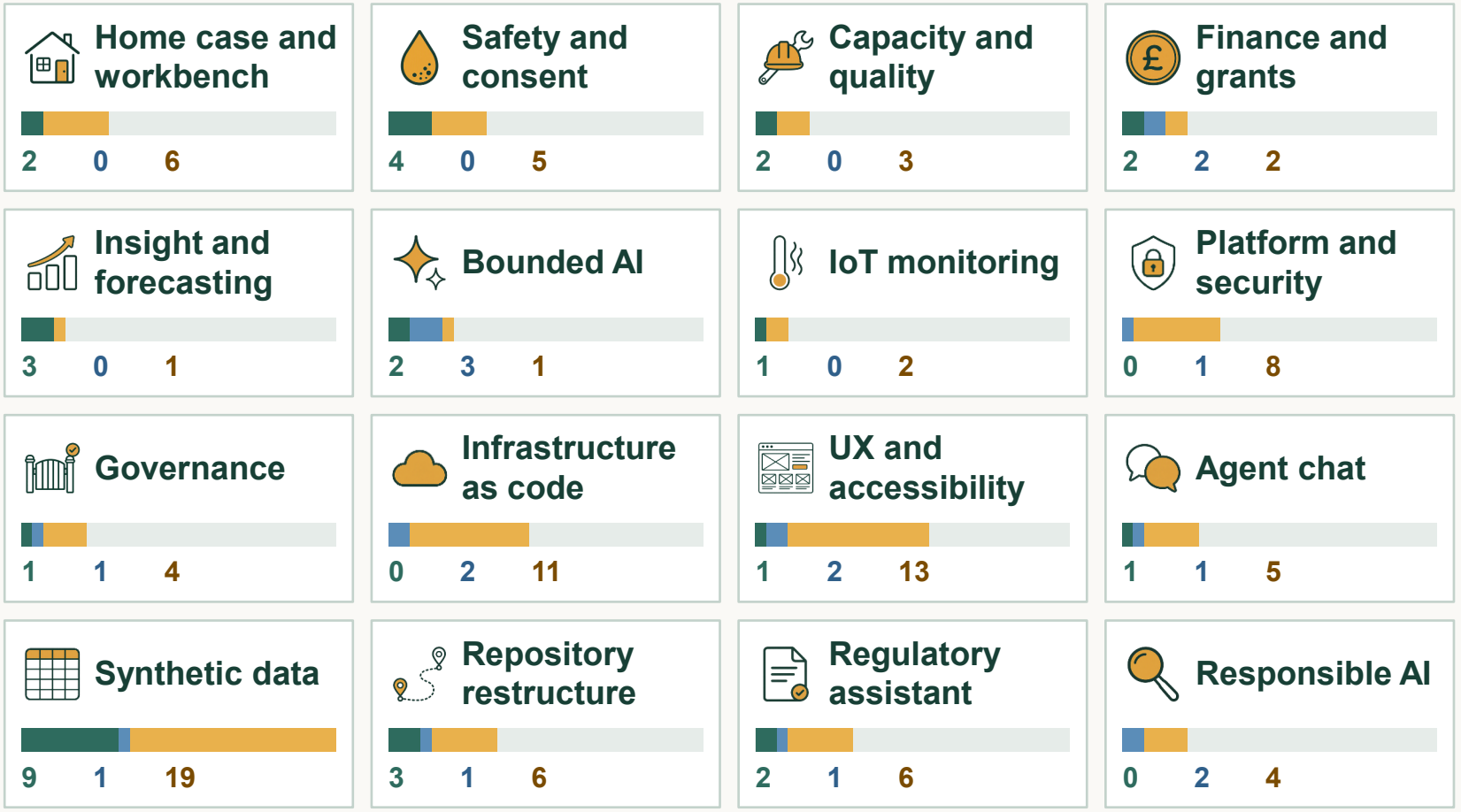
CHANGE EXCEPTION **Changed scope -> named receiver -> affected reviews reopen -> resident update**

Hazards use an independent human route. Access, silence or a booking never implies works consent.

Sources: journey atlas and ADR 0003 (E08) · AMA Plan board: #84, #85, #87, #88, #90, #91, #95, #106

S12 · Answers: B06 · B23

189 backlog items, 16 epics, one coherent product.



54 delivered

24 in progress

111 queued

Generated from the BRD, PRD and architecture decisions

Every item traceable from requirement to pull request

Board totals include 28 items outside the 16 epic areas.
Bars to one scale: full track = 29 items.

Source: AMA Plan project 7 snapshot, 23 Oct 2026 07:21 UTC (API projection of GitHub issues)

Start Monday with the spreadsheets you already run.

Your spreadsheets today

- stock list
- survey tracker
- grant tracker
- contractor schedule

Import preview

Import today's trackers: preview, validate, quarantine bad rows, never touch the source

Map columns · 500-row import under 10 s preview target

DESIGNED · PRD FR-029

Parallel run

Run side by side with named Excel editors, then switch the workbooks to read-only

Read-only comparison

NEXT · #80

Workbench and exports

Export to Excel or CSV any time

Excel or CSV export, no write-back

DESIGNED · FR-030

Progress packs

DELIVERED #102

Evidence export

BUILDING #100

Microsoft 365 for sign-in, alerts and documents

Entra ID sign-in and roles	Microsoft Graph profile	Teams operations alerts	SharePoint evidence library	Teams review nudges
MVP (HIST.)	MVP (HIST.)	CONFIGURED	DESIGNED	ROADMAP

Microsoft 365 is for people and documents; the workbench stays the system of record.

Historical MVP, 12-14 Oct; current staff sign-in/Graph acceptance outstanding.

Sources: PRD FR-029, FR-030, NFR-010 and transition plan · infra/main.bicep · architecture README · AMA Plan #80, #100, #102

S14 · Answers: B20

AI where it earns its place: five assistants, one workbench, people decide.

Rules for policy and arithmetic · learned scoring for candidate choices · generative AI for drafts

Rules

Known policy and arithmetic
Deterministic, fast and repeatable

Decision model

Chooses between supplied candidates
Scores them and can abstain

Generative AI

Explains evidence, drafts letters
Writes a cited draft

Five assistants on one proposal service: prioritise, assess, fund, coordinate, check quality

BUILDING #107

Prioritisation | Assessment | Funding evidence | Coordination | Quality evidence

One proposal service · workflow-only fallback

DELIVERED #105 #106

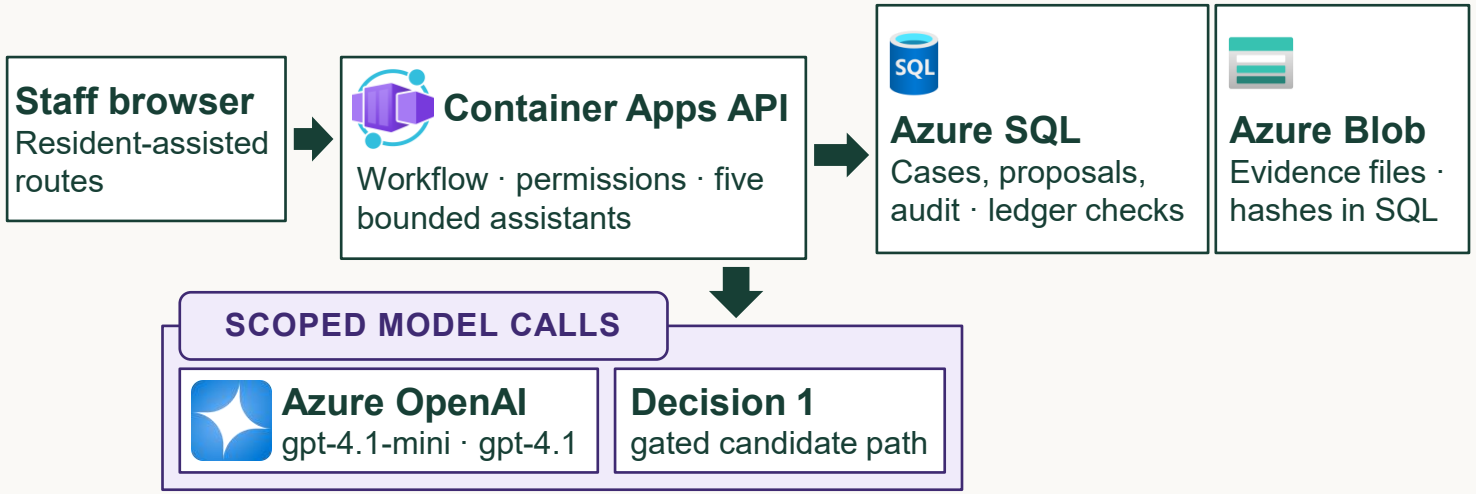


Accept / reject with reason

Every suggestion is accepted or rejected with a reason

Sources: AI placement view (E04, E09) · AMA Plan #104, #105, #106, #107, #214

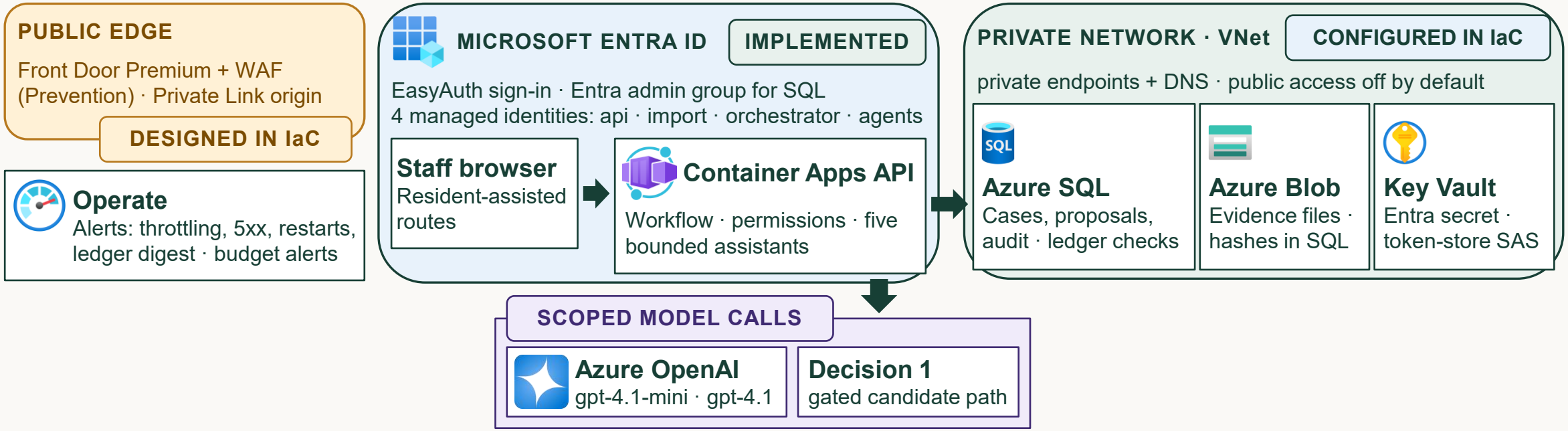
Secure by design: every home record sits inside three boundaries.



Sources: infra/main.bicep · architecture README · ADR 0002 decision 6 · PRD NFR-003 · BRD no-self-approval rules · AMA Plan #80, #102, #143

Filled arrow = request path · tinted region = security boundary · chip = delivery status

Secure by design: every home record sits inside three boundaries.



Current staff sign-in and Graph acceptance remain outstanding.

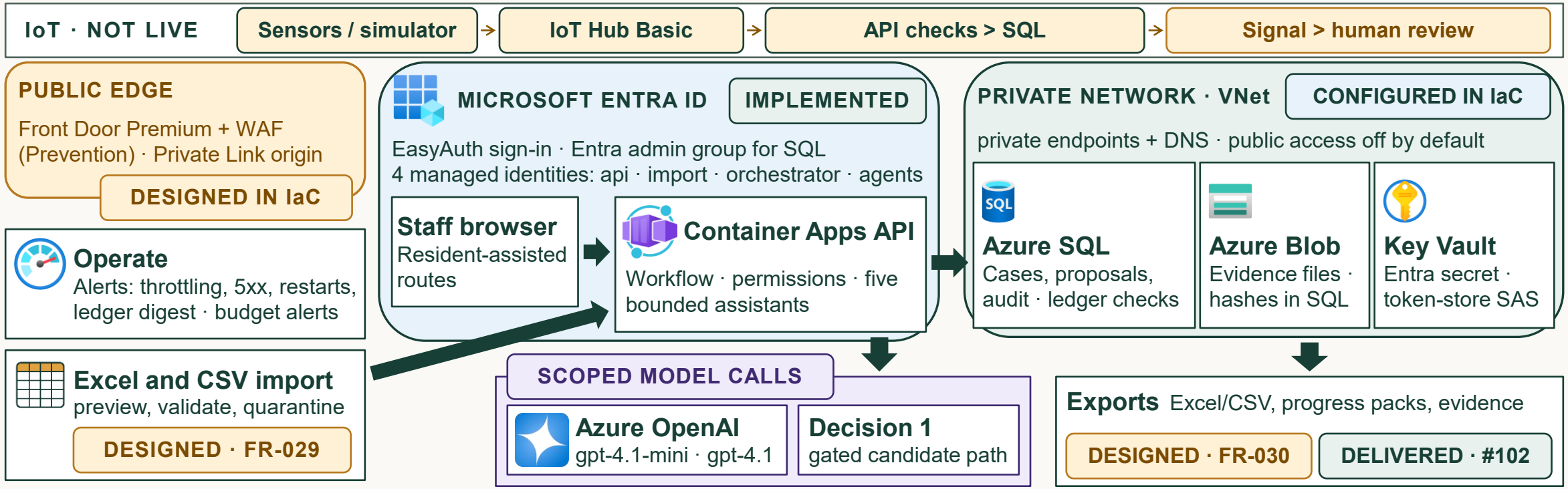
Network: SQL and Key Vault on private endpoints · managed identities · no SQL logins

Sources: infra/main.bicep · architecture README · ADR 0002 decision 6 · PRD NFR-003 · BRD no-self-approval rules · AMA Plan #80, #102, #143

S16 · Answers: B10 · B19 · B22

Filled arrow = request path · tinted region = security boundary · chip = delivery status

Secure by design: every home record sits inside three boundaries.

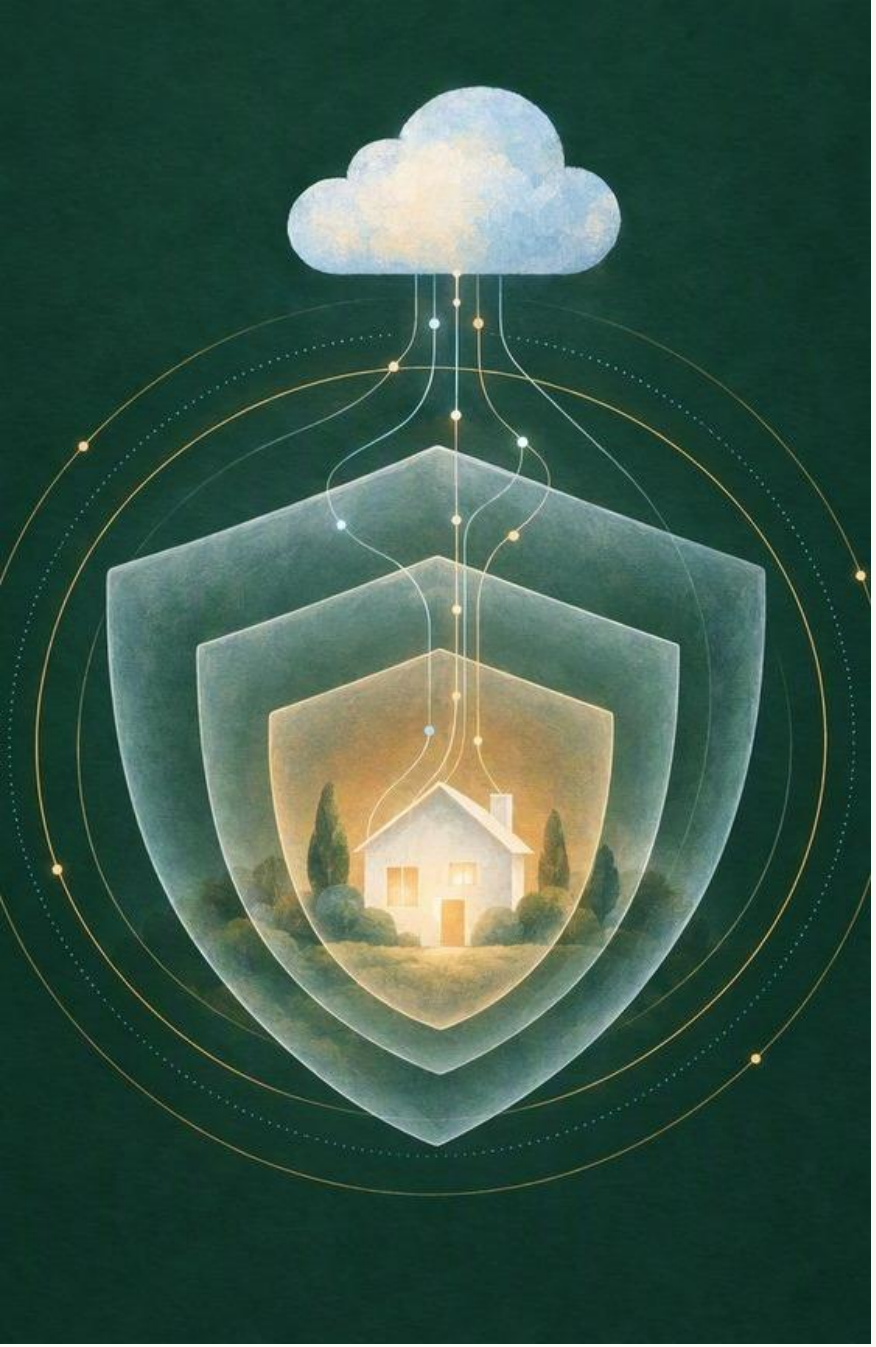


Current staff sign-in and Graph acceptance remain outstanding.

Network: SQL and Key Vault on private endpoints · managed identities · no SQL logins

IoT: monitor after retrofit; no diagnosis. Excel/M365 stay. Portal: assess #441.

Sources: ADR 0002/0003 · infra/main.bicep · PRD NFR-003 · #441 (assessment) · AMA Plan #80, #102, #143



ACT 4 · BUILT TO BE TRUSTED

Seven checkpoints between an AI draft and a real-world decision.

The model prepares. The system checks. A person decides.

1. Signed-in identity (Entra)
2. Role and case scope
3. Permitted evidence only, with source and version IDs
4. Bounded model call, finite allowance and expiry
5. Schema, citation and freshness checks: stale or error stops
6. Human review with current permissions: accept or reject with a reason
7. Separate domain decision and audit trail (consent, quality, Finance)

Failure never becomes approval: the work falls back to its owner
Hazards bypass the AI path entirely
Unknown, stale, disabled or over-allowance → owned manual work · hazards go straight to people

Sources: controlled AI flow (diagram 04, E09) · ADR 0002 · AMA Plan #105, #106

S17 · Answers: B07 · B16 · B21

ACT 4 · BUILT TO BE TRUSTED

Model inference. Measure accepted value.

INFERENCE PLANNING SCENARIO

USD 0.44/home

USD 6,873 / 15,600 homes

20 tasks x 13,768 tokens
x USD 1.60/million

Accepted cost

Unknown

No accepted-outcome
unit cost established

Include review effort
and assurance

Double workload

USD 0.88/home

Inference only. Full project TCO and accepted-outcome cost remain unknown.

IN PLACE

finite calls, tokens and expiry per authority
usage captured on the assistant path
Azure budget alerts

NEXT

proposed money budgets per task and period
per-request call and depth limits
accepted-outcome cost: not yet measured

E21: assumed workload; secondary price, 23 Oct 2026, not current verified price. E19: usage only. E05: controls proposed; #435 not accepted.

S18 · Answers: B05 · B25 · B26 · B27

Everything But the Software

Author retrospective / exploratory single case / non-causal

183 merged PRs

15-24 Oct 2026

36,755 log words

added 15-22 Oct 2026

THE OUTCOME THAT CHALLENGED THE OUTPUT

The author could no longer use the replacement.

Output counts do not establish usable delivery or causation.

Feature -> Tests -> Use journey -> Keep / cut / tidy

Bounded paper test: can another agent supply reflection? Paused; registration pending; no results.

Current source: Everything But the Software, draft + author clarification, 25 Oct 2026; Table 2 defines each window.

ACT 5 · THE PLAN

Validate, pilot, scale: on the clock to April 2030.

30 Nov 2026
Awaab's Law phase 2

31 Mar 2028
wave 3 spend deadline

1 Apr 2030
EPC C deadline

Phase 0
6-8 weeks

Validate one connected journey against a paired comparator

Pilot
6-9 months

TARGETS

- 50+ accepted handovers
- −20% cycle time
- −25% coordination hours per home

Scale
to April 2030

88

homes a week
to reach 15,600 by 1 April 2030

Sources: BRD BG-011 pilot proposal · measurement design · Awaab's Law collection
WH:SHF wave 3 guidance · SRS MEES response

Decision today: fund the validation gate.



DECISION

NEXT GATE

SCOPE	**Fund Phase 0: one connected journey and a paired comparator**
OWNERS	Programme, Finance, technical and control reviewers
SUCCESS MEASURES	Accepted handovers per week · preparation and review time · rework · cost per accepted outcome
COST BOUNDARY	A refreshed, finite budget agreed before spend
SCALE RULE	Scale when quality, control and net value clear the bar

Sources: decision design (E02, E10, E11, E13)

See it work: one home, one exception, one accountable next step.

1



Watch the evidence conflict find its owner

2



Watch what AI drafts, and what only a person can approve

3



Watch the record keep the next action



Ten-minute demo • route confirmed at preflight: live exercise or recorded walkthrough

Sources: demo runbook (E07, E13)

ZAVA FORWARD

More homes ready, with control intact.

1. Agree the validation gate
2. Name the owners
3. Start the 2030 run



APPENDIX · NAVIGATION ONLY · END OF THE SPOKEN ROUTE

Questions we expect, and where the answers are.

STRATEGY AND VALUE

B01 · 28 Why not fix the process and hire more people?
B08 · 35 Will this double delivery and capture GBP 40m?
B09 · 36 What is the per-home case and cash exposure?
B17 · 44 When can we go live, and who owns next?
B20 · 47 Why Azure, not M365, Power Platform or buy?

METHOD AND READINESS

B12 · 39 Validated with real residents and staff?
B13 · 40 How did requirements become the backlog?
B15 · 42 What must happen before a pilot decision?

COST AND TOKENOMICS

B04 · 31 How do Decision Models cut cost and latency?
B05 · 32 What does the technology cost today?
B25 · 52 Tokenomics unit: can the AI bill run away?
B26 · 53 Why not cache everything, use the cheapest model?
B27 · 54 What stops chat and agent sprawl?

ARCHITECTURE AND AI

B02 · 29 Replacing or integrating our housing system?
B03 · 30 Why not exactly the five agents requested?
B10 · 37 Why not deploy each agent independently?
B11 · 38 What proves the AI is real and useful?
B14 · 41 Is it agentic if so much is deterministic?

TRUST, RISK AND PEOPLE

B06 · 33 How do residents keep choice, incl. offline?
B07 · 34 What stops humans rubber-stamping AI?
B21 · 48 How do you prevent hallucination and injection?
B22 · 49 Can data leave the UK or train a model?
B23 · 50 How will you know it is fair and accessible?
B24 · 51 How do policy, law and funding stay current?

OPERATIONS AND DEMO

B16 · 43 What if the model, cloud or deployment fails?
B18 · 45 Can you demo a new live action now?
B19 · 46 Sensors: can AI diagnose damp or mould?

Why not fix the process and hire more people instead?

OPTION 1

Manual / process only

Process improvement is the baseline

TRADE-OFF

Additional staff or contractor capacity may still be necessary

STAFFING PLAN SET IN PILOT

OPTION 2

Workflow + rules

Workflow must beat the baseline

TRADE-OFF

If rules and templates deliver the better accepted-task result, retain them

COMPARISON 1 · READY TO RUN

OPTION 3

Workflow + bounded AI

AI must beat the same workflow without AI

TRADE-OFF

If rules and templates win, report the negative AI finding

COMPARISON 2 · READY TO RUN

ALL OPTIONS **Physical delivery capacity still limits throughput**

Sources: E02 · E10

Is this replacing our housing system or integrating with it?

CURRENT DEMONSTRATOR

Controlled workbench

Demonstrator of coordination and evidence contracts

Not a replacement decision

Housing and repairs integration is the next step

INTEGRATION ROADMAP

- **Housing system**
- **Repairs system**
- **Configured or designed M365 path**

Integration scope, authority and migration need separate discovery

PRODUCTION APPRAISAL MUST IDENTIFY

Authoritative systems · field ownership · APIs · event and reconciliation contracts
Security · retention · the manual service route

Why did you not build exactly the five agents we requested?

Five requested responsibilities are preserved

Autonomy is bounded by decision rights and risk

Five assistants ≠ five independent runtimes

	AI PREPARES	PEOPLE DECIDE
Responsibility	**Bounded help (unreviewed proposal)**	**Human decision owner**
Stock prioritisation	Prioritisation rationale	Approved policy and human review
Assessment	Assessment preparation	Qualified reviewer
Funding	Funding evidence	Finance submits
Coordination	Coordination draft	Resident agrees
Quality	Quality evidence	Independent professional accepts

LATER STAGE

One API hosts the current assistants; independent queue workers remain deferred

Sources: E01 · E03 · E06 · E09

How do Decision Models reduce cost and response time?

Bounded candidates may avoid long-form generation

Compare complete tasks, not endpoint prices

Cheaper and faster results are calibrated in the pilot

$$\text{Cost per accepted task} = \frac{\text{inference + hosting + review and correction + retries + assurance}}{\text{accepted tasks above the quality floor}}$$

Approach	Accepted quality	Total staff time	p50/p95 latency	Cost per accepted task	Critical errors
Rules/manual	Pilot metric	Pilot metric	Pilot metric	Pilot metric	Pilot metric
Parser/templates	Pilot metric	Pilot metric	Pilot metric	Pilot metric	Pilot metric
Decision 1	Pilot metric	Pilot metric	Pilot metric	Pilot metric	Pilot metric
Generation	Pilot metric	Pilot metric	Pilot metric	Pilot metric	Pilot metric

MEASURED IN PILOT

HISTORICAL

Prototype timing: workflow-only gates decide in 0.03 ms; an agent drafting step takes a 12.5 s median. Each does a different job.

Sources: E04 · E05 · E15

What does the technology cost today?

USD 700 owner-set October all-in forecast ceiling

Execution gates remain; authority expires 25 October 16:02 UTC

Refresh post-change forecast; outside-group cost remains unknown

Amount	Scope	Class	Status
USD 700	October all-in forecast ceiling, owner decision 22 Oct	Owner-set ceiling	Paid tranches blocked; expires 25 Oct 16:02 UTC
USD 242.36	Accrued 11-22 Oct, resource group ama	Observed, partial	Estimated until invoiced
USD 535.33	Pre-change scenario	Historical forecast	Superseded for authorisation; refresh
USD 450/month	EXA-012, 10 Oct	Historical allocation	Not current authority
Outside-group spend	Outside resource group	Unknown	Never zero by assumption

DATED LEDGER
NOT PRODUCTION TCO

Future project appraisal
P + USD 6,872.9856
Per planned home:
(P + USD 6,872.9856) / 15,600
P = non-inference costs for the same period.
P is not yet priced.

Count each cost once. Do not add the October USD 700 envelope and its included inference.

Sources: E11 / E05 / E21. Historical price assumption: 23 Oct 2026; no verified current quote.

How do residents retain choice, especially without digital access?



Optional portal; assisted routes stay

Offer ≠ agreement; silence ≠ consent

Representative authority is task-limited

DESIGN BOUNDARY · HUMAN VALIDATION NEXT

Direct resident research, approved communications and privacy decisions remain necessary

Sources: E02 · E08 · E12

What stops a human from rubber-stamping AI?

Source verification before proposal acceptance

Current permission, evidence and independent authority

Meaningful oversight remains a human acceptance gate

1. Capture

Role + case scope
Permitted evidence
Source/version IDs

AI PREPARES

2. Call

Bounded task
Model + usage
No database lock

3. Validate

Schema + citations
Stale/error = stop
Store as unreviewed

PEOPLE DECIDE

4. Review

Human checks source
Current permissions

Separate domain action

Failure does not become approval

Unknown, stale, disabled, exhausted allowance or unavailable model → owned manual work; hazards bypass AI.

Citation presence is not semantic truth · Proposal review is not consent, QA or Finance

SOURCE VIEW · 23 OCT 2026

Sources: E02 · E09 · E12

Will this double retrofit delivery and capture GBP 40 million?

1 · AMBITION

Original brief ambitions: the outcomes the pilot measures

Double retrofit delivery

Capture GBP 40 million

ORIGINAL BRIEF AMBITION

2 · DEPENDENCIES

Capacity, scope, eligibility and cash all constrain value

Survey, technical, contractor, material, resident and QA capacity

Grant availability and secured cash: confirmed at the funding gate

3 · TEST

Frozen cohorts and attributable comparisons required

Measure accepted handovers and the whole cost of the work

READY TO RUN

4 · OBSERVED RESULT

NEXT MILESTONE

Benefit range set once the baseline, causal limits and sensitivity are agreed

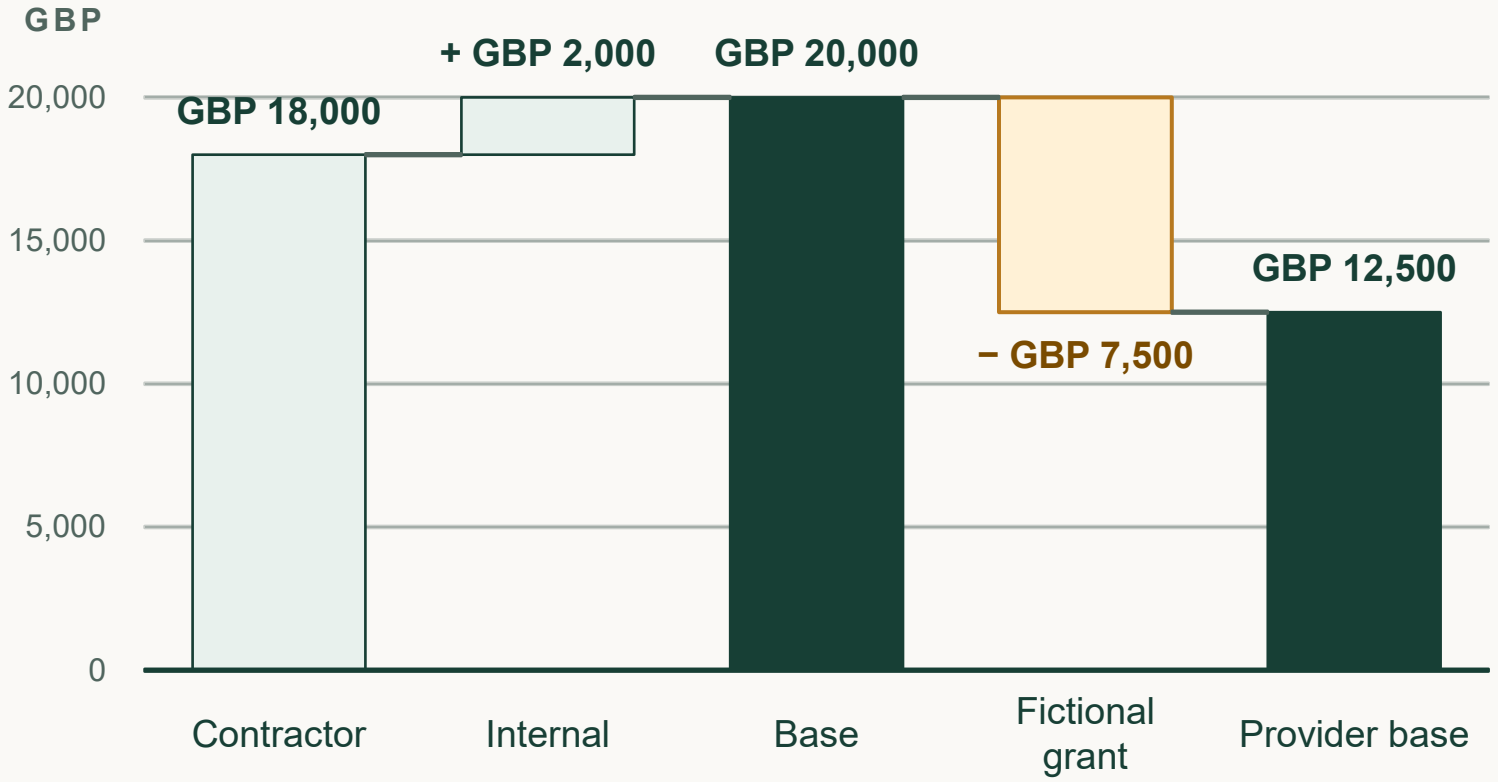
Saved minutes become cash through a Finance-approved mechanism

Sources: E01 · E02 · E10

What is the per-home business case and cash exposure?

Benchmark: GBP 6,082/home
x 15,600 = GBP 94,879,200 works

Conditional: min(50% x GBP 6,082, GBP 7,500)
= GBP 3,041/home; x 15,600 = GBP 47,439,600



FICTIONAL EXERCISE ARITHMETIC

Fictional base GBP 20,000
GBP 22,000 with contingency

Gross cash GBP 20,000 before grant
Fictional receivable GBP 7,500
An award is not cash.

Contingency spent: provider GBP 14,500
Evidence supports claims; not eligibility, award or payment.

E02 / E21: historical benchmark, not quote. Grant assumes eligible homes/costs, allocation, award and claim; Finance/funder review required.

Why not deploy each agent independently?

CURRENT RUNTIME

Current runtime: shared API and SQL

One API · five bounded assistants

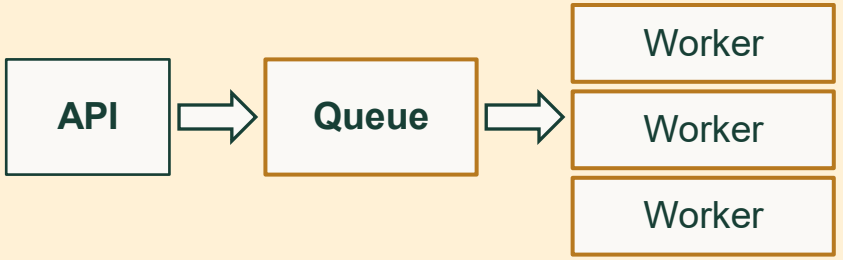
Prioritisation | Assessment | Funding | Coordination | Quality

SQL state
Short SQL transactions · inference outside locks

FUTURE SPLIT

LATER STAGE

Separate workers add failure and operating boundaries



They also introduce queue ownership, retries, duplicate handling and distributed state.

WHEN TO SPLIT

Split when measured isolation or load justifies it
A future split needs explicit contracts, operational evidence and a separate decision.

Filled arrow = current application/state relationship · Hollow arrow = roadmap path

Sources: E06 · E09 · E13

What proves the AI is real and useful rather than a template?

DATED HISTORICAL EVIDENCE

Invocation
Provider/model/usage and source=ai evidence
Historical real calls and persistence are documented.

≠

NEXT MILESTONE

Semantic acceptance
Semantic quality, critical omissions, unsafe advice and fairness require separate evaluation.

NEXT MILESTONE

Incremental value
Review burden and incremental value require separate evaluation.

LABELLED FALLBACK

Template/replay is clearly labelled as a fallback
The real-inference gate runs on live model calls.

WHAT THE PILOT ADDS

Real invocation proven; calibrated quality and value come from the pilot
Live readiness is re-checked before every run, never assumed.

Sources: E07 · E09 · E04 · E13

Did you validate this with real residents and staff?

No direct resident or staff research has been completed for this design.

Representative five-stage maps / pain hypotheses / desired gains, not measured outcomes

Resident	**1 Hear / report**	**2 Survey**	**3 Choose works**	**4 Appointment**	**5 Live with measures**
	Hypothesis: repeat story / pressure. Gain sought: meaningful choice + owned aftercare.				
Assessor	**1 Prepare**	**2 Arrange access**	**3 Observe**	**4 Write up**	**5 Clarify / revisit**
	Hypothesis: missing records / re-keying. Gain sought: a sourced, traceable assessment.				
Finance preparer	**1 Establish need**	**2 Check route**	**3 Assemble case**	**4 Maintain claim**	**5 Reconcile**
	Hypothesis: conflicting versions / unsupported scope. Gain sought: a defensible pack.				

Next: direct task/comprehension tests, assisted routes, real handoffs and net effort. No endorsement.

Current source: journey atlas, 14 Oct 2026 research / 23 Oct reconciliation; representative hypotheses, not user validation.

How did the requirements become an agent-delivered backlog?

CONCRETE BACKLOG TRACE / retained 23 Oct 2026 deck snapshot

BRD / PRD	Epic #73	Feature #76	Story #77	PR #376

PR evidence is not whole-journey acceptance. HVE has human stops; the coordinator was owner-designed.

Paper's bounded test: can another agent supply reflection between features?

A / Feature only

No extra review budget

B / Consolidate

Review / refactor + tests / note

C / Read-only review

Same extra budget as B

Planned S1: 4 runs per arm / 10 features + CR1. Exploratory case, not causal proof.

PAUSED / registration pending / no results. B/C have extra time vs A; tests and notes also differ.

Current source: Everything But the Software, draft + author clarification, 25 Oct 2026. Historical trace retained above.

Sources: E14 · E16 · E17 · E20

Is this actually agentic if so much is deterministic?

Policy and state

RULES · NO AI

Deterministic orchestration protects controlled state
The application determines eligibility and protects state.

Reasoning and preparation

AI PREPARES

Reasoning and preparation are bounded capabilities
Models reason over permitted context and prepare drafts or flags.

Accountable decision

PEOPLE DECIDE

People retain decisions requiring consent, professional judgement or delegation.

AUTONOMY

Autonomy is task-specific, not a marketing label
Future autonomous actions require an explicit, evaluated task contract and authorised boundary.

Sources: E03 · E06 · E09

What exactly must happen before a pilot or production decision?

NEXT GATE

Current deployment and AI evidence

Current deployment, roles, SQL and real-AI evidence

OWNER ROLE

Technical owner

REQUIRED ARTIFACT

Staff access and role denial, task-bound real inference, SQL persistence on the selected deployment

NEXT GATE

Human and control acceptance

Human, privacy, professional and accessibility acceptance

OWNER ROLE

Professional, privacy, accessibility and safety reviewers

REQUIRED ARTIFACT

Task/comprehension, accessibility, safety/privacy and professional acceptance

NEXT GATE

Value, recovery and release

Measured value, cost, recovery and explicit release decision

OWNER ROLE

Programme and Finance

REQUIRED ARTIFACT

Measured value and cost against the comparator, recovery test, explicit release decision

Sources: E02 · E12 · E13 · E17

What happens if the model, cloud or deployment fails?

Resilience is configured in the architecture; whole-system recovery drills are the next milestone.



Access + secrets

Entra and managed identities
Key Vault secret references
Private backing services



Runtime + releases

Container Apps / pinned images
Registry and CI checks
Controlled rollout + rollback



Observe + budget

Monitor / Log Analytics
Scoped allowance + usage
Budget alerts warn; hard stops are designed.

Evidence required before expansion

NEXT GATE

Genuine staff sign-in + role denial	Restore and recovery drill
Azure SQL persistence	Human task/comprehension checks
Bounded real-model result	Current cost + processing terms

Restart proven; full restore drill is the next milestone

OWNED MANUAL ROUTE

No approval from failure or stale evidence
Owned manual work and independent safety handling
Keep manual and safety routes independent; do not relax private controls to rescue a demo.

Columns are concerns, not request hops · Amber = next milestone

SOURCE VIEW · 23 OCT 2026

Sources: E06 · E07 · E13

When can we go live and who owns the next step?

Dependency-led plan: dates follow the gates

SEQUENCED PLAN

READINESS SEQUENCE	OWNER	
1 Agreed validation scope	Programme	Programme owns scope; Finance owns spend review
2 Refreshed cost	Finance	
3 Deployment prerequisites	Technical owner	Technical and control owners supply acceptance evidence
4 Finite test authorised	Programme + Finance	
5 Dates set	→	LATER, SEPARATE DECISIONS: Commercial pilot; Operational release

Can you demonstrate a new live action right now?

DEMO BOUNDARY

Only pre-authorised, reversible constructed scope
No unplanned role, network, spend or data changes
We will not grant a role, bypass private networking, enable assistant authority, submit a grant or send a resident message on the fly.

DEMO ROUTE, CONFIRMED AT PREFLIGHT

→ **L · Live, preflight passed**
Only within the preflighted, authorised demonstration boundary

→ **E · Recorded walkthrough**
Fallback is visibly labelled evidence, never simulated live proof
Switch once; say exactly what it does and does not prove.

→ **U · Unavailable**
Explain the control instead

× **No silent mock fallback**

Sources: E07 · E13

Where do sensors fit and can AI diagnose damp or mould?

IoT Hub Basic is an exercise design decision

Offline consumer tested; live Hub path remains roadmap

Signals request human review, never diagnose or suppress

TARGET DESIGN · ROADMAP

Accepted design direction; the live telemetry path is on the roadmap.

Device / simulator

Distinct identity
Label synthetic input
Stable reading ID

Azure IoT Hub

Basic for exercise
Device-to-cloud only
Credential owner needed

API + SQL

Validate mapping/time
Idempotent append
Owned DQ exceptions

Human review

Deterministic signal
No diagnosis / repriority
Missing = insufficient

Independent safety route

A resident report goes directly to human hazard handling. Telemetry and AI never delay or suppress it.

Amber box + hollow arrow = target design on the roadmap

Sources: E12

Why Azure instead of M365, Power Platform or buying a product?

Appraise buy/configure/build on the same requirements

The Azure exercise proves the contracts end to end

Licensing, integration, control and TCO decide

Option	Workflow and permissions	Audit and integration	Accessibility	Model terms	Total cost
Existing housing product (buy)	Evidence required	Evidence required	Evidence required	Evidence required	Evidence required
M365 / Power Platform (configure)	Evidence required	Evidence required	Evidence required	Evidence required	Evidence required
Custom Azure (build: the exercise realisation)	Evidence required	Evidence required	Evidence required	Evidence required	Evidence required

PROCUREMENT DECISION: NEXT STEP

Licences and tenant capability are confirmed in Phase 0.

The platform choice follows customer value.

Sources: E02 · E03 · E06

How do you prevent hallucinations and prompt injection?

Permitted task context and source/version binding

Schema/citation checks plus independent semantic review

Built to contain vulnerabilities and hallucinations, not to deny them

1. Capture

Role + case scope
Permitted evidence
Source/version IDs

2. Call

Bounded task
Model + usage
No database lock

3. Validate

Schema + citations
Stale/error = stop
Store as unreviewed

4. Review

Human checks source
Current permissions
Separate domain action

UNTRUSTED EVIDENCE

Documents are evidence, not instructions

REVIEW BOUNDARY

Citation presence is not semantic truth

Failure does not become approval

Unknown, stale, disabled, exhausted allowance or unavailable model → owned manual work; hazards bypass AI.

Filled arrow = guarded request/response step, not agent-to-agent talk

SOURCE VIEW · 23 OCT 2026

Sources: E09 · E12 · E07

Can customer data leave the UK or train a provider's model?

Processing geography is set by the deployment type

UK-only inference is a deployment choice (a regional deployment rather than GlobalStandard), confirmed at the security gate.

Verify exact model, deployment type and contract

before real data is processed

- VERIFY Provider
- VERIFY Model and version
- VERIFY Deployment SKU
- VERIFY Routing and geography
- VERIFY Retention
- VERIFY Abuse monitoring
- VERIFY Subprocessors
- VERIFY Applicable agreement

No real resident data or private dataset in this handoff

The handoff contains only public synthetic illustrations and repository-authored summaries, not private datasets or credentials.

How will you know this is fair and accessible?

No-digital route and understandable explanations

Cohort errors, exclusions and unknowns are visible

Engineering checks ≠ resident or WCAG acceptance

	Task success	Comprehension	Errors and exclusions by cohort
Digital route	MEASURED IN PILOT	MEASURED IN PILOT	MEASURED IN PILOT
No-digital route	MEASURED IN PILOT	MEASURED IN PILOT	MEASURED IN PILOT

Inspect errors and exclusions by agreed cohorts without exposing small groups

GUARDRAIL

An AI result that improves average speed while worsening a critical group outcome should not pass the gate.

Conformance and fairness are assessed with real users and a full audit in the pilot.

Sources: E02 · E12 · E15

How will you keep policy, law and funding rules current?

Detected change → reviewed rule change → tests

REPOSITORY CONVENTION · READY TO ACTIVATE

Source

Dated sources with authority classification

Reviewed change item

Affected requirements, rules, journeys, tests

NAMED OWNER

Approved control version

Human approval of a reviewed control update

NAMED OWNER

Regression evidence

Tests and evidence before consequential use

Models may retrieve, compare and flag

They must not decide that a legal duty is satisfied

No silent model update to live controls

Or silently activate a new rule

Dated official-source snapshots; currentness and named legal review required before consequential use.

Sources: E02 · E12 · E18

What is your Tokenomics unit and can the AI bill run away?

Planning: USD 0.440576/home inference; accepted-outcome TCA unknown

Historical secondary price: 23 Oct 2026. Assumed workload, not current verified deployment price.

Observe, then assume

6,884 / 5 = 1,376.8 tokens/run
x 10 = 13,768 tokens/task
x 20 = 275,360 tokens/home

Price inference only

275,360 / 1m x USD 1.60
= USD 0.440576/home
x 15,600 = USD 6,872.9856

Measure accepted TCA

Hosting + review + assurance
Accepted above quality floor
TCA / accepted work: unknown
Double workload: USD 0.881152/home

Inference scenario is not total project cost or accepted value

DESIGNED · NEXT MILESTONE

Existing bounded path

Finite calls / tokens / expiry; partial metering
Not hard currency or request-depth limits.

Proposed: not run

Complete attribution; task/period money stops
Request/depth bounds; billing reconciliation.

Tokenomics framing: our working method, refined with Finance

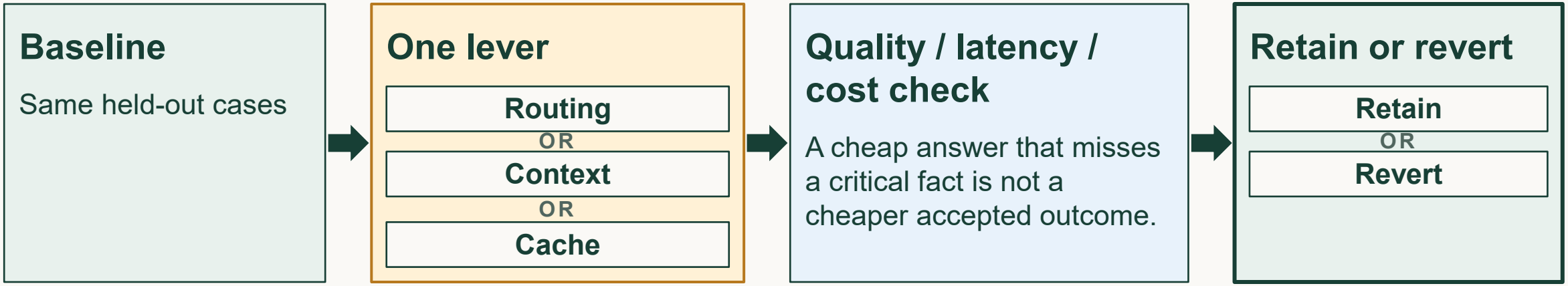
RECOVERY: 24 OCT 2026

Sources E19 / E05 / E11 / E21. #435 not accepted; no saving, payback or full-project price established.

Why not cache every answer and route everything to the cheapest model?

Change one lever, compare, then keep or revert

OPTIMISATIONS · ON THE ROADMAP



CACHE

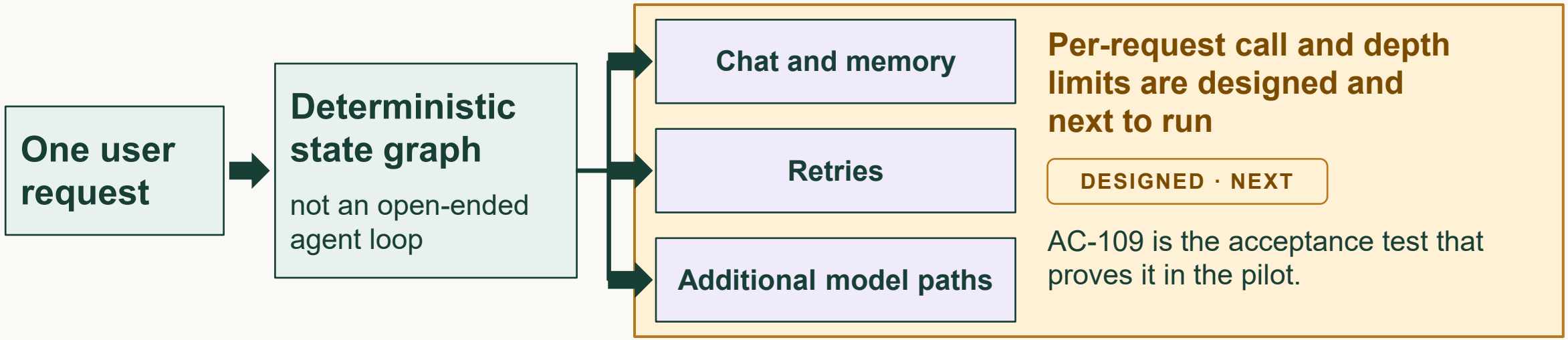
Same permission, evidence version and task validity required

ROUTING

Smallest model must clear held-out quality and latency thresholds

Sources: E19 · E04 · E09

What stops chat and agent sprawl multiplying hidden calls?



Existing bounded authority expires and exhausts

Finite calls, tokens and expiry per authority
Money caps come next

DESIGNED · NEXT

Complete per-call metering, request attribution, task and period money budgets, and a workflow-only stop

Amber box with outline = designed bound, next milestone · Purple = bounded model call · Green = existing workflow and authority

Sources: E19 · E09